SAAT Limits, Derivatives, Definite Integrals & Area

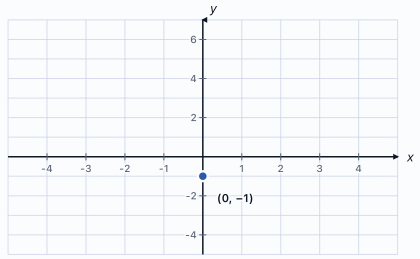
PRACTICE WORKSHEET

25 Structured Exam-Style Questions **Name:** _____ **Date:** _____ **Score:** _____ / 25

SAAT Limits, Derivatives, Definite Integrals & Area Practice Exercises

1 MCQ

The coordinate diagram shows the parabolic curve $f_1(x) = x^2 - 20x + 118$ in the domain of SAAT Differential & Integral Calculus. By completing the square $f_1(x) = (x - h)^2 + k$, find the sum $k + f_1(12)$.



A 50
B 22
C 38
D 40

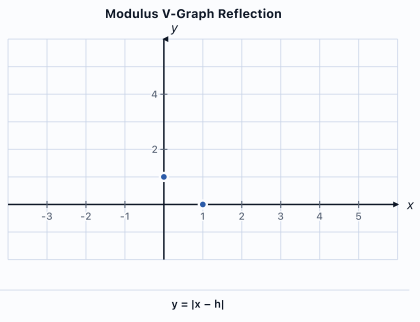
2 MCQ

A mathematical model in SAAT Differential & Integral Calculus is defined by the cubic function $y_2(x) = 3x^3 - 10x^2 + 15x - 15$. Calculate the exact gradient of the tangent to the curve at the point where $x = 3$.

A 36
B 96
C 21
D 22

3 MCQ

In SAAT Differential & Integral Calculus, two displacement vectors are given by $\mathbf{u}_3 = \langle 10, -8 \rangle$ and $\mathbf{v}_3 = \langle 9, 7 \rangle$. Compute the exact scalar value of $\mathbf{u}_3 \cdot \mathbf{v}_3 + \|\mathbf{u}_3\|^2$.



A 34
B 198
C 164
D 310

4 MCQ

The linear growth profile shown in the diagram for SAAT Differential & Integral Calculus follows an arithmetic progression with first term $a_1 = 11$ and common difference $d = 6$. Find the exact sum S_{15} of the first 15 terms.

A 95

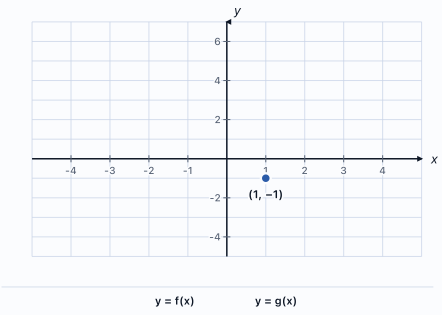
B 885

C 795

D 1425

5 MCQ

In SAAT Differential & Integral Calculus, evaluate the exact definite integral $I_5 = \int_0^2 (18x^2 + 40x)\,\mathrm{d}x$.



A 152

B 168

C 65

D 128

6 MCQ

In SAAT Differential & Integral Calculus, consider the quadratic function $f_6(x) = x^2 - 22x + 211$. Express $f_6(x)$ in the completed-square form $(x-h)^2 + k$ and compute the exact value of $k + f_6(18)$.

A 229

B 240

C 139

D 222

7 MCQ

The graph illustrates the cubic model $y_7(x) = 3x^3 - 11x^2 + 33x - 22$ studied in SAAT Differential & Integral Calculus. Find the instantaneous gradient $\frac{\mathrm{d}y}{\mathrm{d}x}$ of the tangent to the curve at $x = 8$.

A 609

B 433

C 400

D 83

8 MCQ

In SAAT Differential & Integral Calculus, two displacement vectors are given by $\mathbf{u}_8 = \langle 11, -7 \rangle$ and $\mathbf{v}_8 = \langle 8, 5 \rangle$. Compute the exact scalar value of $\mathbf{u}_8 \cdot \mathbf{v}_8 + \|\mathbf{u}_8\|^2$.

A 53

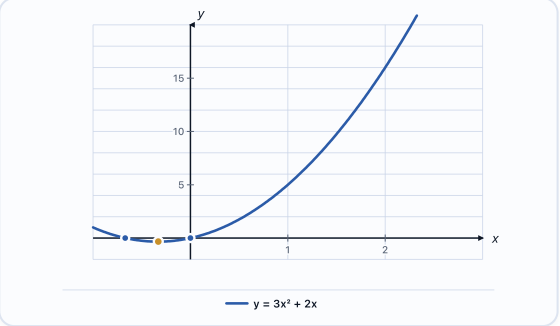
B 170

C 223

D 293

9 MCQ

Evaluate the definite integral $\int_0^2 (3x^2 + 2x)\,dx$.



A 10

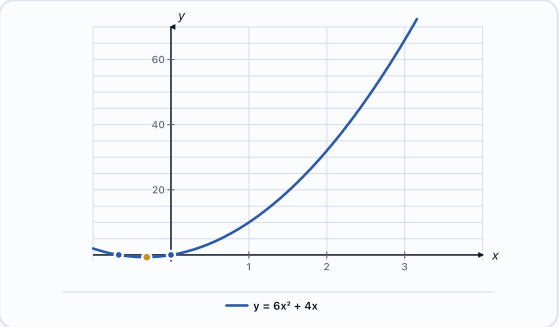
B 12

C 14

D 16

10 MCQ

Evaluate the definite integral $\int_0^3 (6x^2 + 4x)\,dx$.



A 66

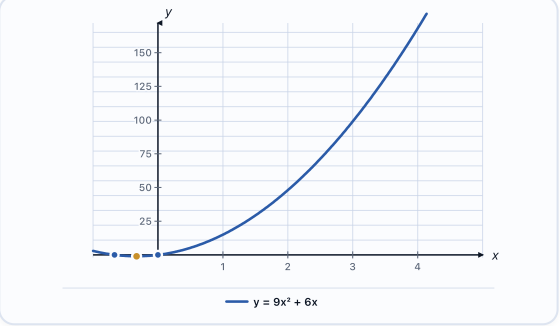
B 69

C 72

D 76

11 MCQ

Evaluate the definite integral $\int_0^4 (9x^2 + 6x)\,dx$.



A 168

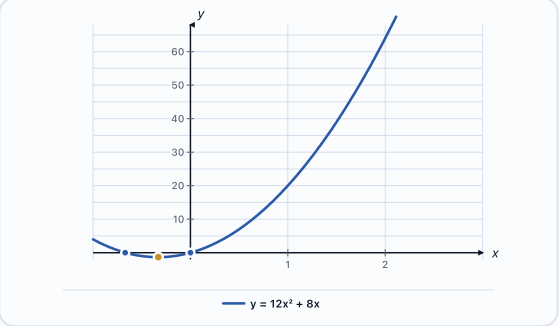
B 240

C 246

D 236

12 MCQ

Evaluate the definite integral $\int_0^2 (12x^2 + 8x)\,dx$.



A 56

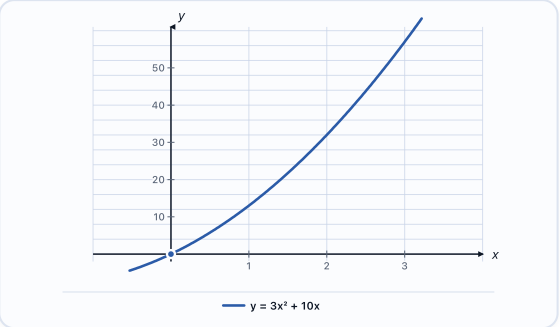
B 64

C 46

D 48

13 MCQ

Evaluate the definite integral $\int_0^3 (3x^2 + 10x)\,dx$.



A 82

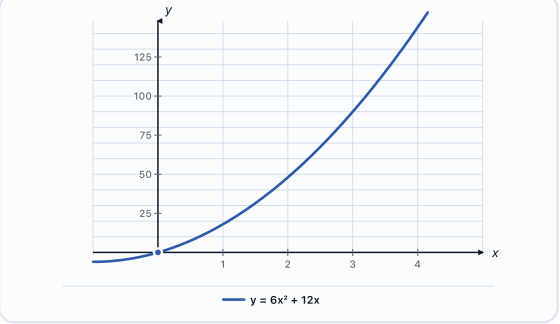
B 69

C 72

D 57

14 MCQ

Evaluate the definite integral $\int_0^4 (6x^2 + 12x)\,dx$.



A 220

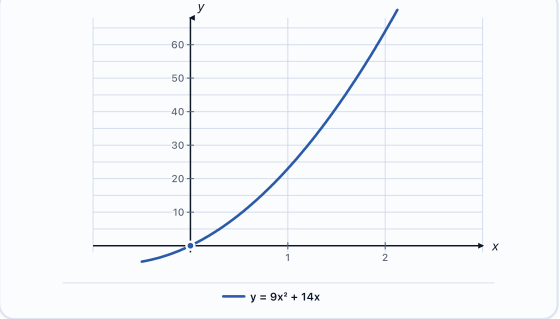
B 144

C 224

D 236

15 MCQ

Evaluate the definite integral $\int_0^2 (9x^2 + 14x)\,dx$.



A 64

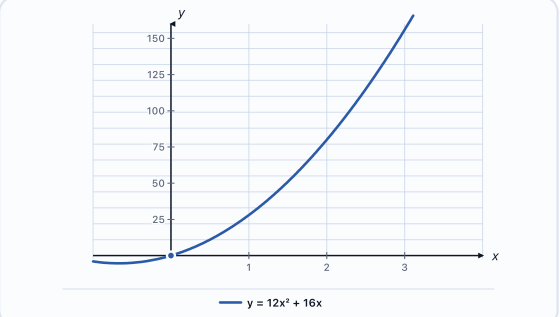
B 50

C 66

D 52

16 MCQ

Evaluate the definite integral $\int_0^3 (12x^2 + 16x)\,dx$.



A 180

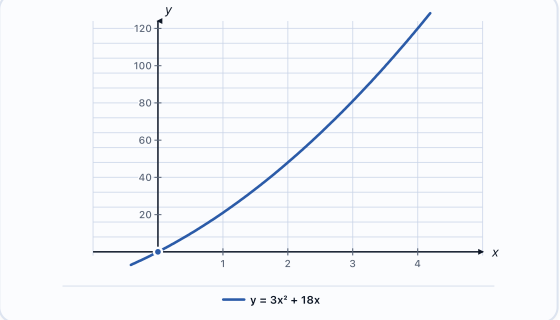
B 196

C 177

D 156

17 MCQ

Evaluate the definite integral $\int_0^4 (3x^2 + 18x)\,dx$.



A 120

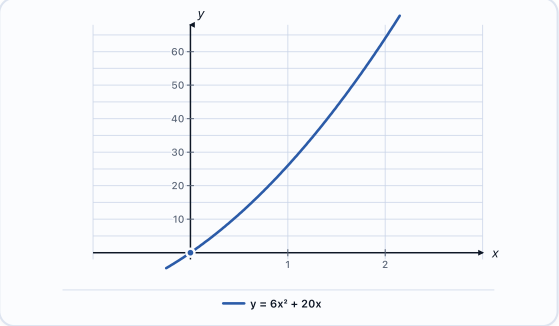
B 226

C 204

D 208

18 MCQ

Evaluate the definite integral $\int_0^2 (6x^2 + 20x)\,dx$.



A 56

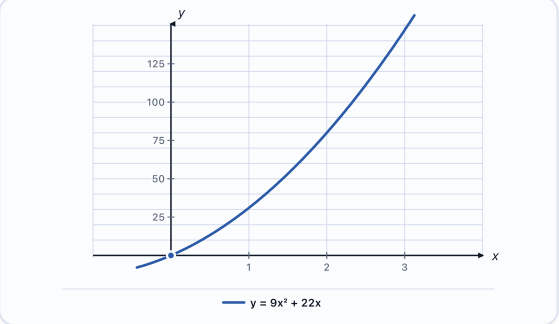
B 64

C 76

D 54

19 MCQ

Evaluate the definite integral $\int_0^3 (9x^2 + 22x)\,dx$.



A 177

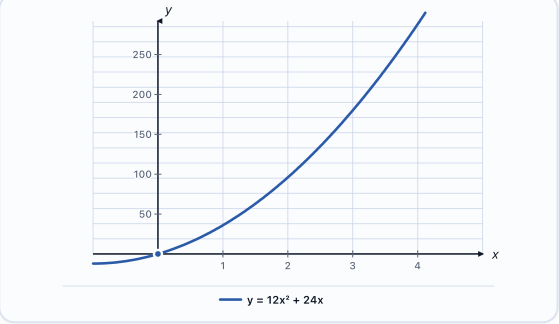
B 202

C 147

D 180

20 MCQ

Evaluate the definite integral $\int_0^4 (12x^2 + 24x)\,dx$.



A 444

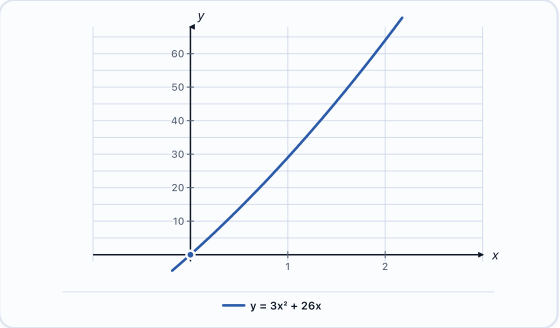
B 472

C 288

D 448

21 MCQ

Evaluate the definite integral $\int_0^2 (3x^2 + 26x)\,dx$.



A 86

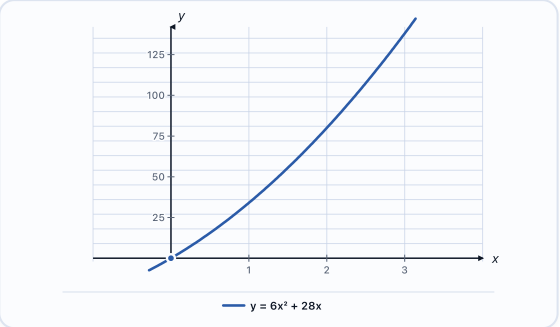
B 64

C 58

D 60

22 MCQ

Evaluate the definite integral $\int_0^3 (6x^2 + 28x)\,dx$.



A 180

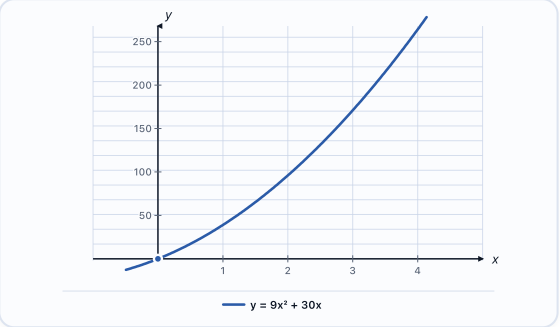
B 177

C 138

D 208

23 MCQ

Evaluate the definite integral $\int_0^4 (9x^2 + 30x)\,dx$.



A 264

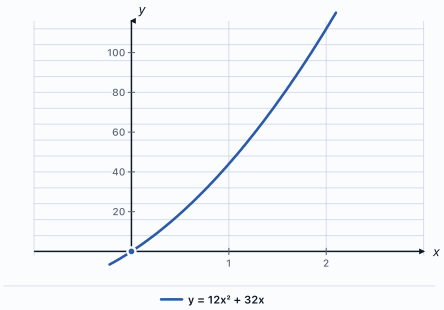
B 432

C 462

D 428

24 MCQ

Evaluate the definite integral $\int_0^2 (12x^2 + 32x)\,dx$.



A 94

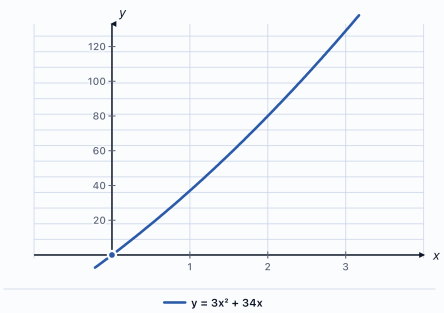
B 128

C 96

D 112

25 MCQ

Evaluate the definite integral $\int_0^3 (3x^2 + 34x)\,dx$.



A 129

B 180

C 214

D 177